



From lecture session 4: We realized that continental margins (shelf, slope, rise) and abyssal plain are all covered in sediments. Here we will explore what sediments are and what are the sources of sediments to the ocean.

## Lecture Session 5

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1. Sediment classifications: size and source;
2. Sources, transport, and environmental factors;
3. Studying sediments: methods and sediments as resource.

Somewhere in the Caribbean

## Icebreaker

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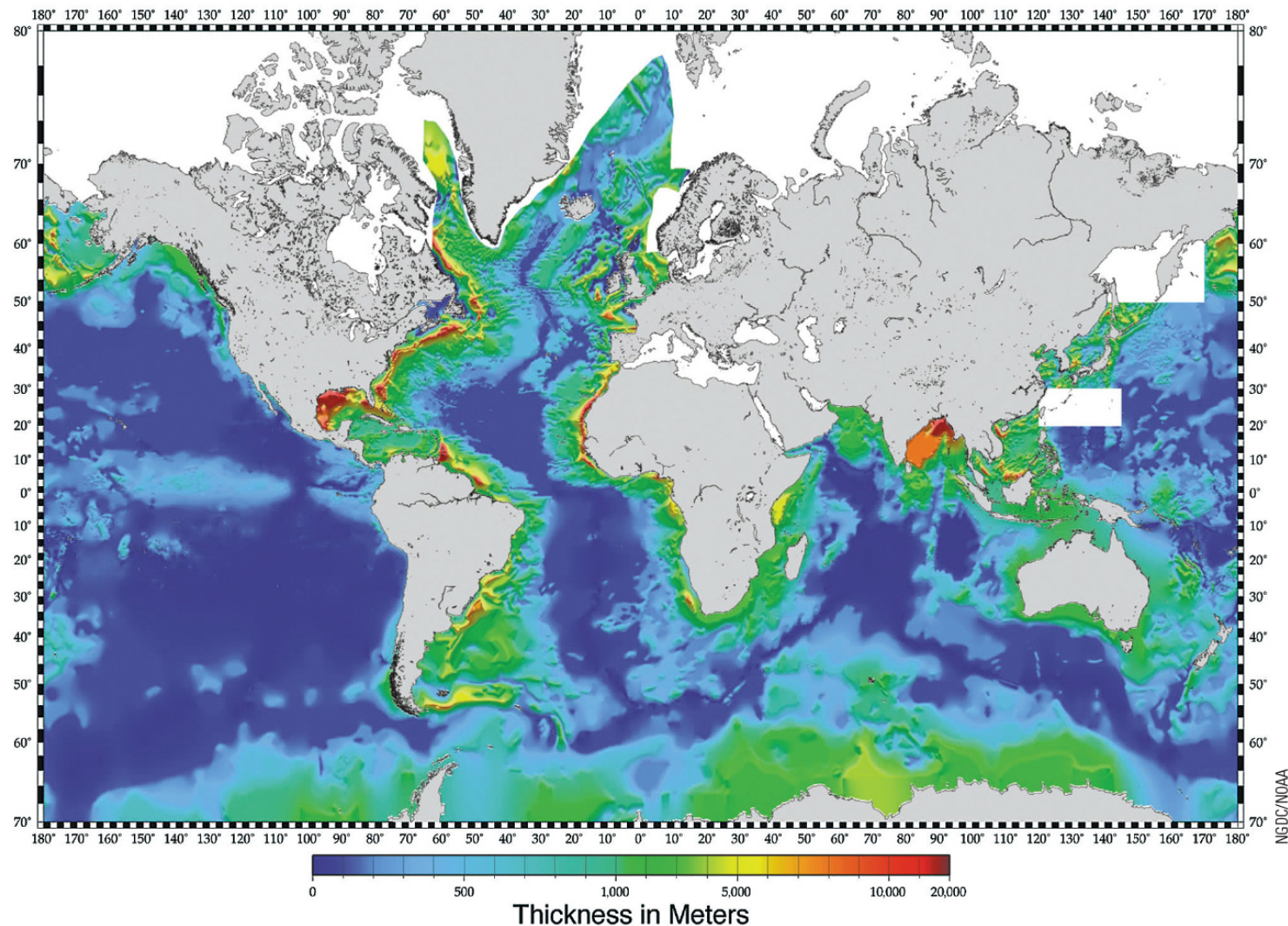
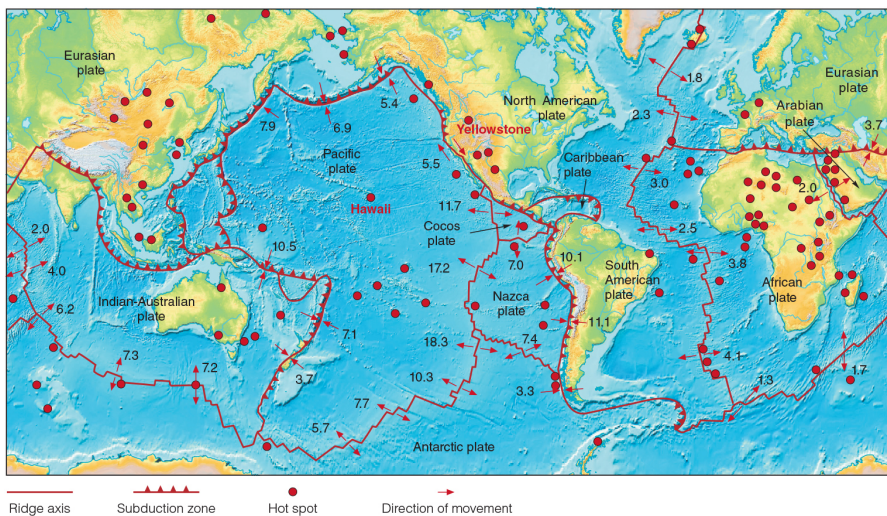
1. What are sediments?
2. Are sediments the same everywhere? Based on your experience what are the differences between the “material” covering the bottom of a lake, bay, river, or beach you know? Provide details: like size, color, odor, texture, etc.
3. Based on your previous answers, can you create a classification method for these sediments? Why do they present such characteristics?



# The Average Thickness Of The Marine Sediments

## Icebreaker 2

Can you see any correspondence between this map and what you know about plate tectonics and the ocean's different regions?



**Figure 5.8** Total sediment thickness of the ocean floor, with the thinnest deposits in dark blue and the thickest in red. Note the abundant deposits along the east and Gulf coast of North America, in the South China Sea, and in the Bay of Bengal east of India.

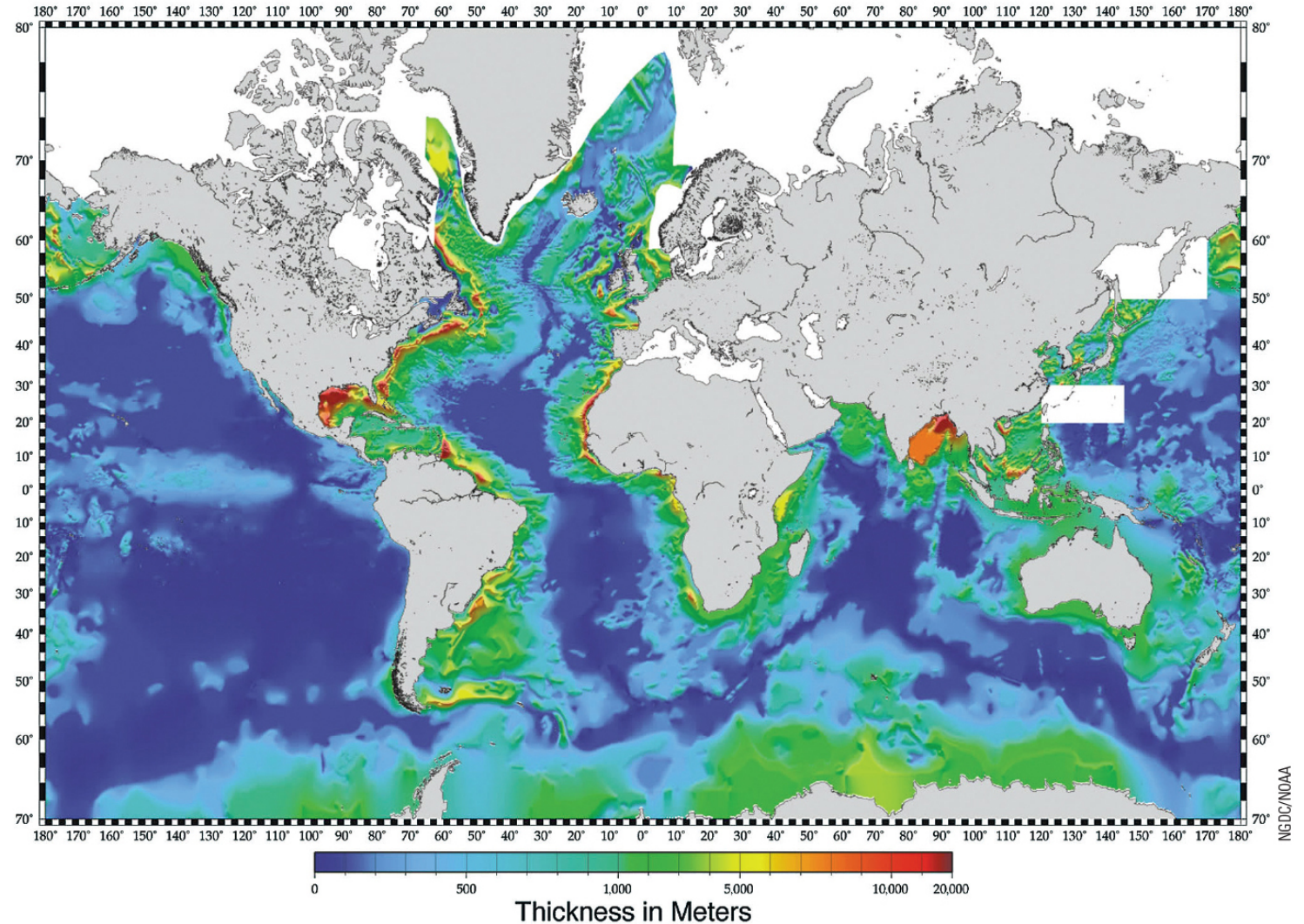
# The Average Thickness Of The Marine Sediments

## Example

Can you see any correspondence between this map and what you know about plate tectonics and the ocean's different regions?

The thickest deposits are located far from the tectonic plate edges and near the continents. Mid-oceanic ridges, trenches, and other volcanic active areas far from the continents have the thinnest deposits.

Source of sediments, concentration of ocean life, and destruction and creation of solid surfaces explain this pattern.

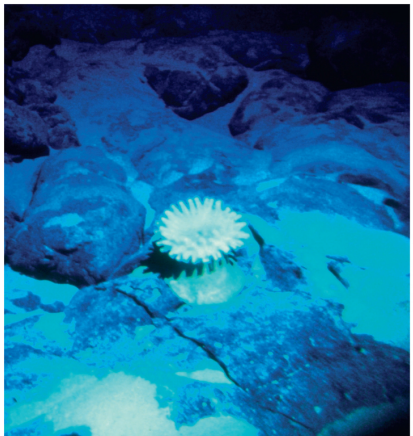


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# Classifying Sediments

**Sediments** = particles of organic or inorganic matter that accumulate in a loose, unconsolidated form

From the icebreaker exercise I hope you realized sediments have a wide range of characteristics. Such characteristics are determined by source, chemical composition, transport history, biological activity, physical and chemical properties of the environment. **Therefore, sediments are considered the memory of the ocean.**



**Figure 5.1** Sediment near the crest of the Mid-Atlantic Ridge. A sea anemone clings to newly formed rock outcrops only lightly dusted with sediment.



**Figure 5.2** Brittle stars and their tracks on the continental slope off the coast of New England. The depth here is 1,476 meters (4,842 feet).



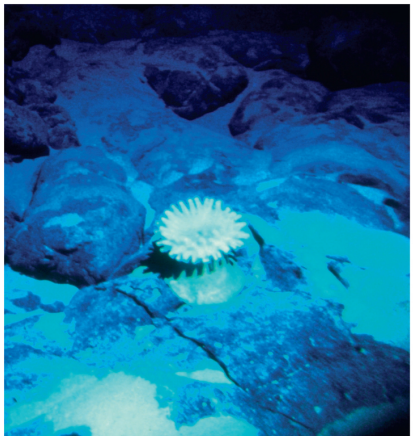
**Figure 5.17** A community of marine organisms live in the sediment near McMurdo Sound, Antarctica.

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Very few places of the ocean are sediment free. Examples: 1- new crust might not been created long enough to be covered by sediments; 2 – Strong currents can scour the sediments away; 3- Chemical reactions can help dissolve sediments faster than deposition rates.



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# Classifying Sediments: Particle Size

**Table 5.1** Particle Sizes and Settling Rate in Sediment

|        | Size Class               | Size             | Size Comparison            | Settling Velocity in Still Water | Time to Settle<br>4 Kilometers (2.5 Miles) |
|--------|--------------------------|------------------|----------------------------|----------------------------------|--------------------------------------------|
| Gravel | Boulder                  | >256 mm          | Basketball                 |                                  |                                            |
|        | Cobble                   | 64–256 mm        | Potato to grapefruit       |                                  |                                            |
|        | Pebble                   | 4–64 mm          | Throwing and skipping size |                                  |                                            |
|        | Granule                  | 2–4 mm           | Pea                        |                                  |                                            |
| Sand   | Very coarse sand         | 1–2 mm           | Peppercorns                |                                  |                                            |
|        | Coarse sand              | 0.5–1 mm         | Coarse sugar               | 2.5 cm/sec (1 inch/sec)          | 1.8 days                                   |
|        | Medium sand              | 0.25–0.5 mm      | Granulated sugar           |                                  |                                            |
|        | Fine sand                | 0.125–0.25 mm    | Confectioners' sugar       |                                  |                                            |
|        | Very fine sand           | 0.0625–0.125 mm  | Visible to the eye         |                                  |                                            |
| Mud    | Coarse silt              | 0.0310–0.0625 mm | Barely visible to the eye  | 0.025 cm/sec (1/100 inch/sec)    | 6 months                                   |
|        | Medium to very fine silt | 0.0039–0.0310 mm | Microscopic                |                                  |                                            |
|        | Clay                     | <0.0039 mm       | Microscopic                | 0.00025 cm/sec                   | 50 years                                   |

Based on the Udden-Wentworth Sediment Grain Size Scale.

# Classifying Sediments: Particle Size

- Most common marine sediments: **sand, silt, clay**. Silts and clays consist what we call mud.
- Generally, the smaller the particle, the more easily it can be transported by streams, waves, and currents.
- As sediment is transported it gets sorted by size.

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- Sediment can contain particles of similar size, or it can be a mixture of different-sized particles. This mixture will depend on the sorting mechanism. Highly energetic phenomena (turbidity currents) and environments will transport several classes of sediments making them poorly sorted.

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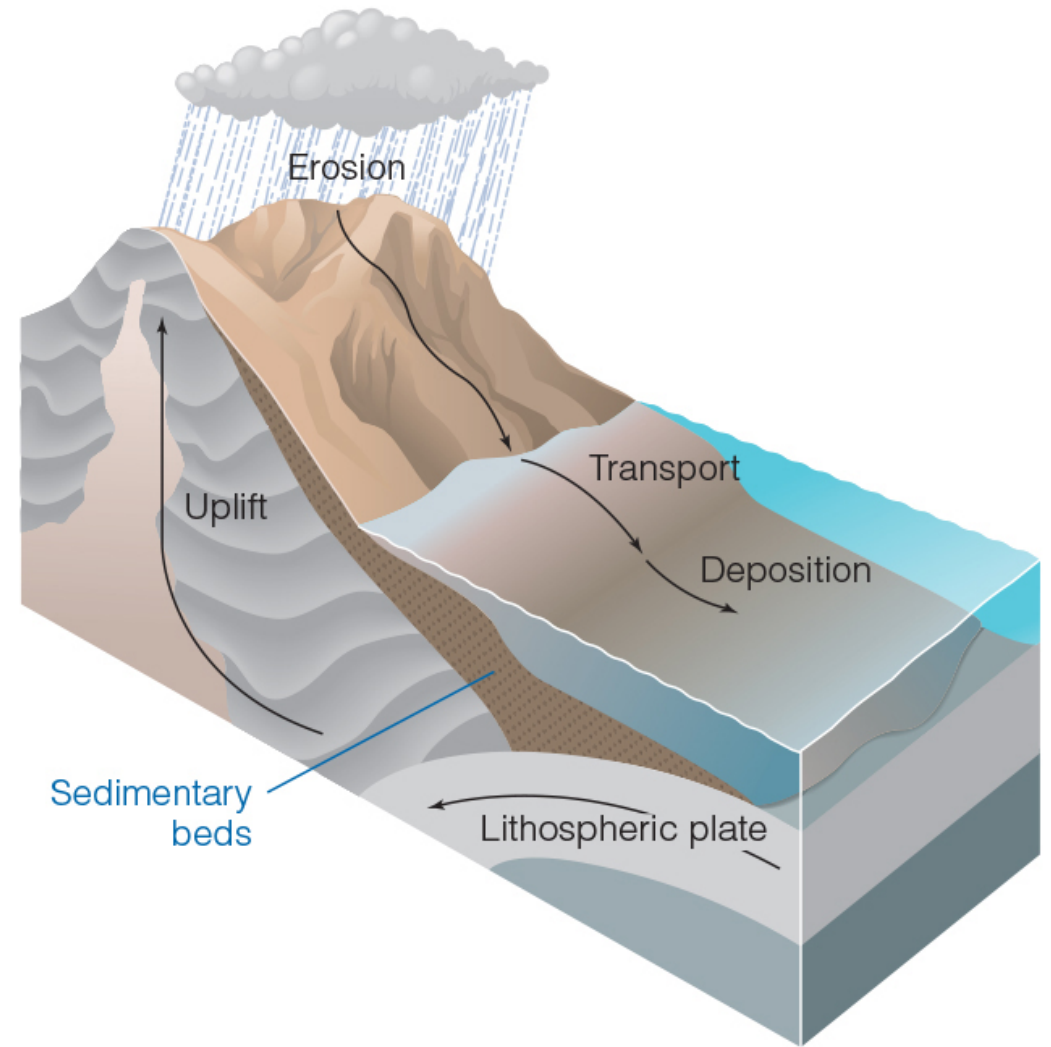
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# Classifying Sediments: Source Type

# Terrigenous

Are mainly transported to the ocean by rivers and streams (~15 billion tons every year) and by winds (~10 million tons every year of fine sediments and volcanic ash).

Covers mostly the continental margins, especially the continental shelves!



**Figure 5.5** A simplified sediment cycle. Over geologic time, mountains rise as lithospheric (crustal) plates collide, fuse, and subduct. Water and wind erode the mountains and transport resulting sediment to the sea. The sediments are deposited on the seafloor, where they travel with the plate and are either uplifted or subducted. The material is made into mountains again.

# Terrigenous

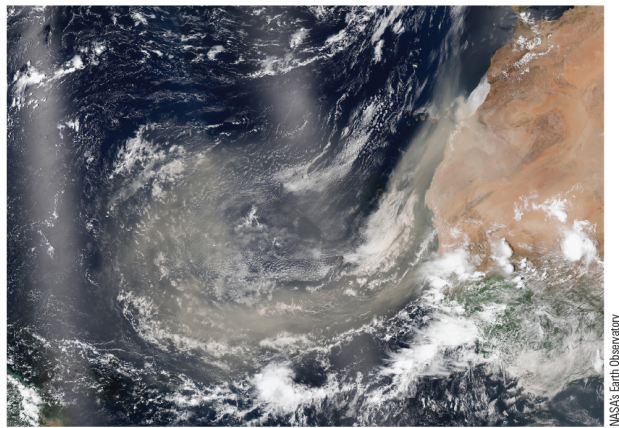
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- a** Rivers are the main source of terrigenous sediments. This photo, taken from space, shows sediment entering the Gulf of Mexico from the Mississippi River.

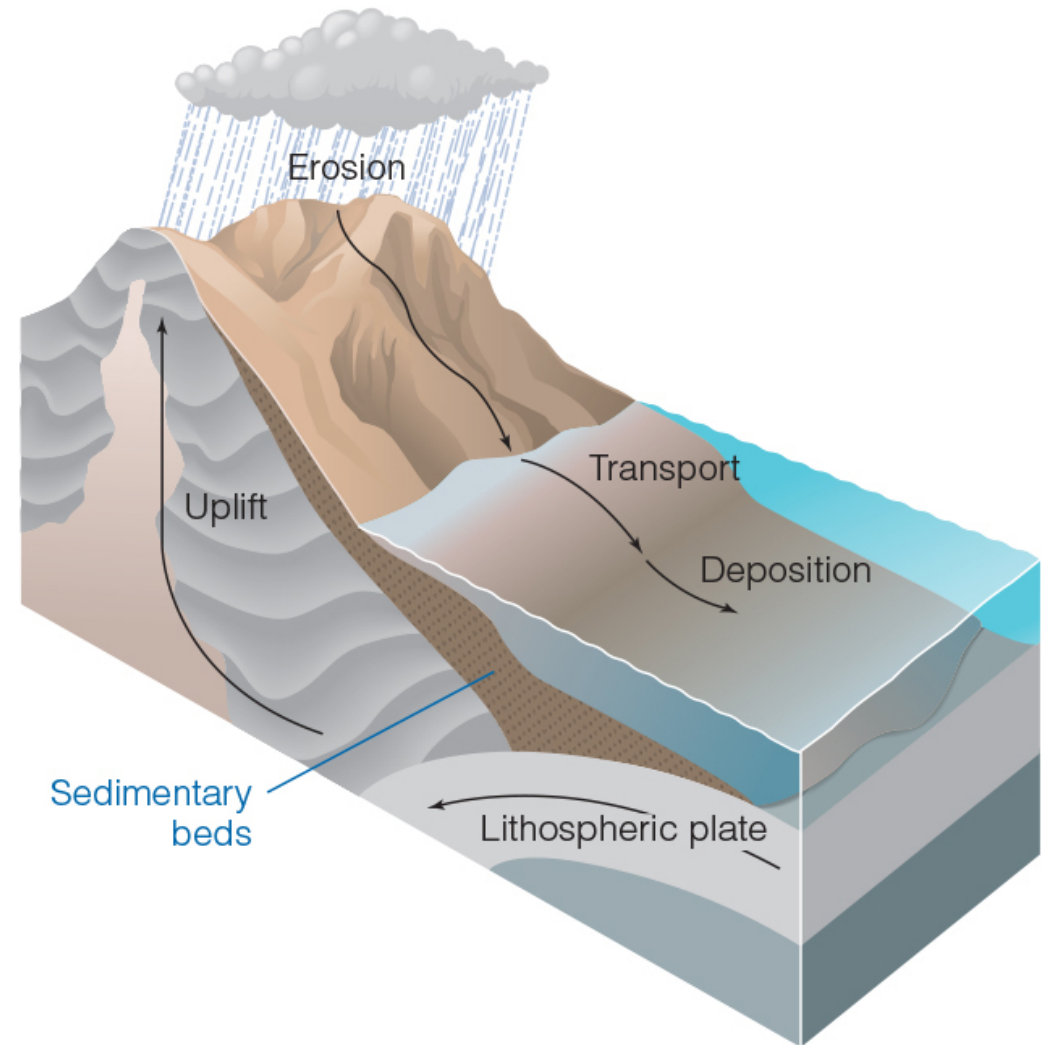
**Figure 5.6** Sources of terrigenous sediments.



- b** Dust from the Gobi Desert blows eastward across the Pacific on 18 March 2002. The particles will fall to the ocean surface and descend slowly to the bottom to end up as terrigenous sediments.



- c** The ash cloud caused by the April 2010 eruption of Eyjafjallajökull, a volcano in Iceland. The wind transported ash from the eruption for hundreds of kilometers and greatly disrupted aviation in Europe. Much of the ash ended up in the north Atlantic.



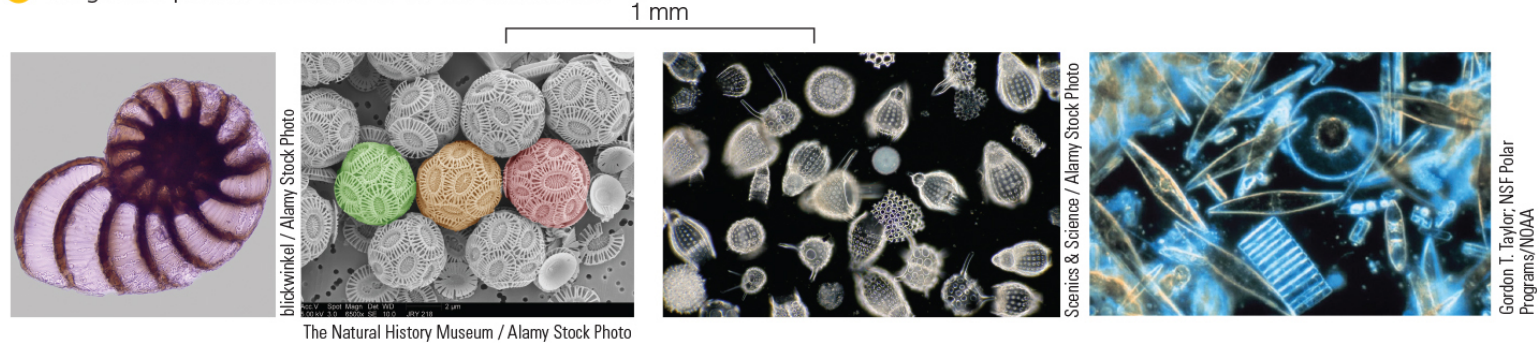
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# Biogenous: sediments generated by marine life. They consist of inorganic or organic matter.

- Although the volume and mass of Terrigenous sediments are larger, biogenous sediments cover a larger area. **Can you imagine why?**

- Inorganic Sediments (most abundant biogenous sediments):** Mainly from calcareous and siliceous materials from organisms' shells. After they die, their hard parts sink and accumulate at the bottom

**a** The general pattern of sediments on the ocean floor.



**b** A foraminiferan, an amoeba-like animal.

**Figure 5.9** Ocean sediment distribution.

**c** Coccoliths, individual plates of coccolithophores, a form of planktonic algae. Because of their tendency to dissolve, calcareous oozes very rarely occur at bottom depths below 4,500 meters (14,800 feet). Coccolithophores are much smaller than the other ooze-forming organisms shown in these figures.

**d** Radiolaria, another siliceous ooze-former. Radiolarian oozes are most common in regions of equatorial upwelling.

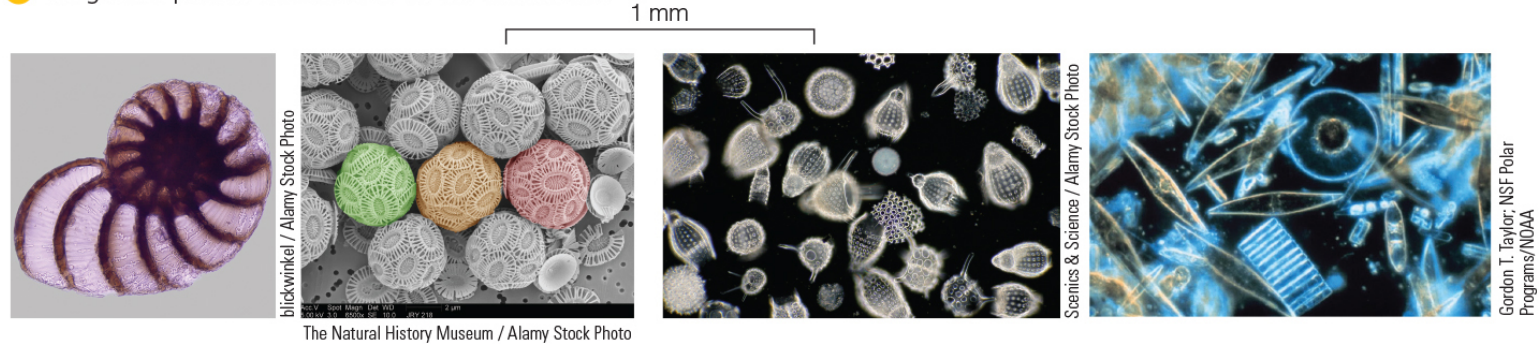
**e** Shells of diatoms, single-celled algae. Diatom oozes are most common at high latitudes and in areas of upwelling.

Ooze = deep-ocean sediment containing at least 30% biogenous material.

# Biogenous (sediments generated by marine life)

- Although the volume and mass of Terrigenous sediments are larger, biogenous sediments cover a larger area. **Can you imagine why?**  
**Marine life are present everywhere!**
  - More abundant in areas of high biological activity. Example: Coastal Upwelling Systems (reason why SD waters are so cold!)
  - Cover most of the deep ocean areas
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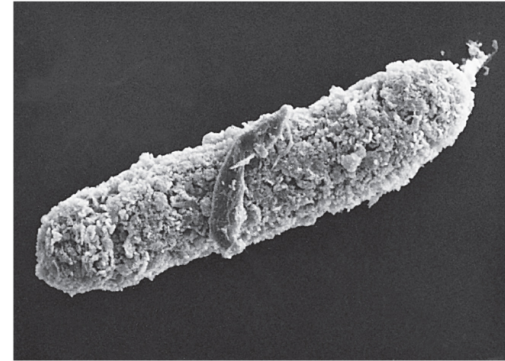
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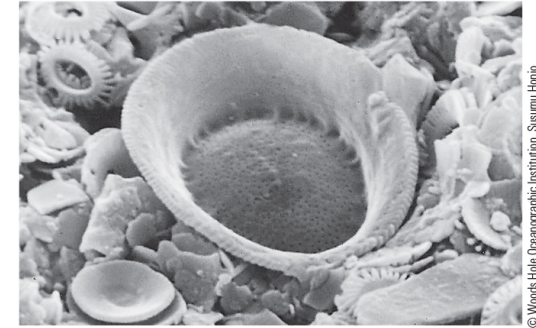
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- **Organic Sediments:** Fecal matter and dying organisms. Not as abundant than inorganic due to decomposition and other process.



© Woods Hole Oceanographic Institution, Susumu Horiya

**a** A fecal pellet of a small planktonic animal. The compressed pellet is about 80 micrometers long.



© Woods Hole Oceanographic Institution, Susumu Horiya

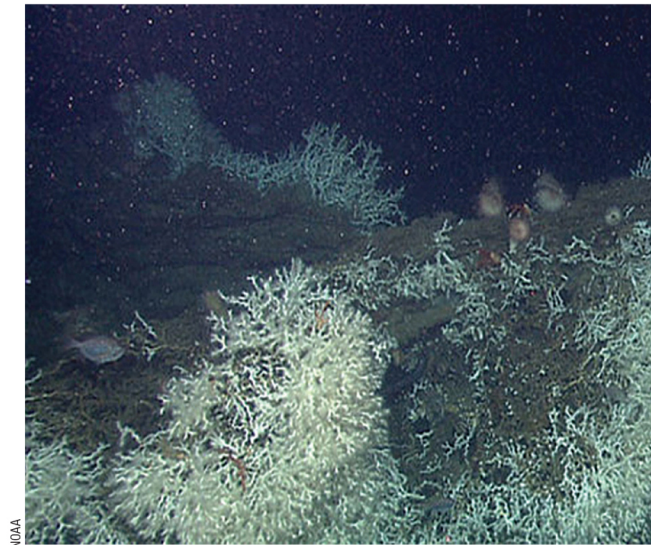
**b** Enlargement of the pellet's surface, magnified about 2,000 times. The pellet consists of the indigestible remains of small microscopic plantlike organisms, mostly coccolithophores. Unaided by pellet packing, these remains might take months to reach the seabed, but compressed in this way, they fall much faster and can be added to the ooze in perhaps two weeks.



Susumu Horiya Woods Hole Oceanographic Institution

**c** Marine snow is an aggregation of organic debris that increases settling rates and provides a source of biological material to seafloor sediments.

**Figure 5.16** Sinking organic debris.

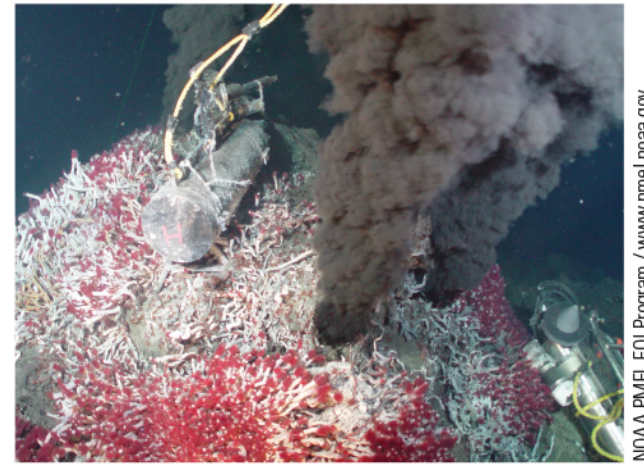


NOAA

**d** Marine snow slowly drifting down onto a shipwreck in the Gulf of Mexico.

# Hydrogenous (Authigenic) Sediments

Precipitation of minerals dissolved in seawater. Very common near hydrothermal vents



**Figure A** An ROV sampling near a hydrothermal vent community along the Galapagos Rift.

## Cosmogenous Sediments

Stuff that comes from outer space!

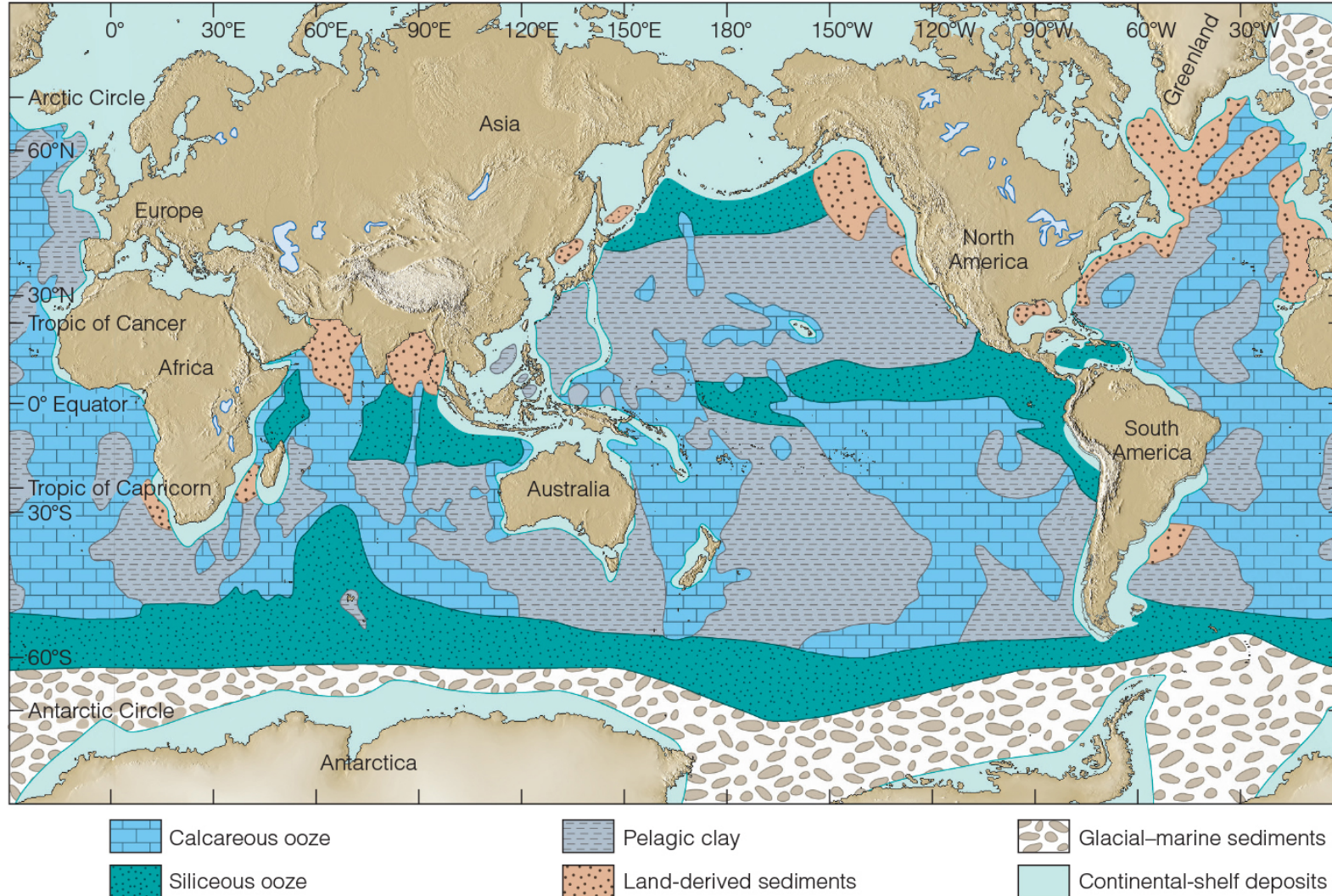
Most of the deposits of are a mixture between terrigenous and biogenous sediments with some supplement of hydrogenous and cosmogenous sediments.

**Figure 5.7** Microtektites, very rare particles that began a long journey when a large body impacted Earth and ejected material from Earth's crust. Some of this material traveled through space, reentered Earth's atmosphere, melted, and took on a rounded or teardrop shape. These specimens of sculptured glass range from 0.2 to 0.8 millimeter in length. Glassy dust much finer in size, as well as nut-sized chunks, have also fallen on Earth.



Michael Daniels

# Distribution of Sediments:



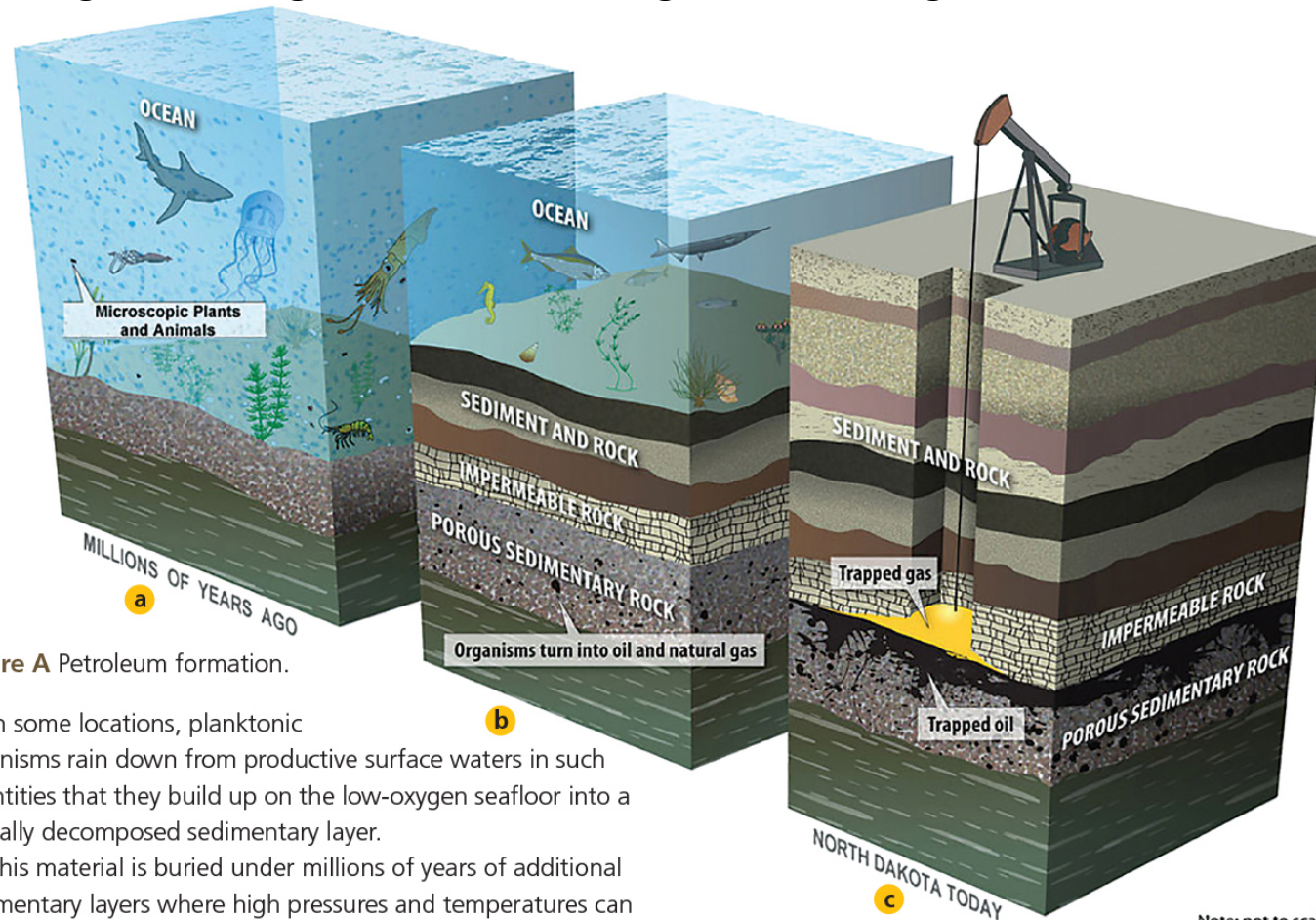
Note that the sediments' distribution patterns are different depending on the oceanic basin.

Each pattern, sediments' size, chemical composition, age, and deposits' thickness contain information about oceanic processes (physical, biological, and chemical), geological history, and economic potential of certain places (**THE HISTORY OF THE OCEAN**).

Marine geologists specialized in sediments are responsible for decoding such information.

# Economic importance of studying ocean sediments

Oil and gas: Biogeneous organic sediments originate oil and gas



**Figure A** Petroleum formation.

- a** In some locations, planktonic organisms rain down from productive surface waters in such quantities that they build up on the low-oxygen seafloor into a partially decomposed sedimentary layer.
- b** This material is buried under millions of years of additional sedimentary layers where high pressures and temperatures can slowly transform it into petroleum.
- c** The relatively low-density petroleum migrates from its source rock up through porous reservoir rocks where it may be trapped by impermeable cap rocks. It can remain there for many millions of years unless found by humans and pumped to the surface to be processed into a myriad of products, including gasoline.

Source: Courtesy, ND Geological Survey, Becky Barnes, artist.

# Economic importance of studying ocean sediments

Coastal erosion: Understanding the sediments movements along coastal areas can help us predict sea level rise damages.



# Preparing for Friday's discussion – Deep Sea Mining (Heffernan, 2019)

<https://www.nature.com/articles/d41586-019-02242-y>

Guiding questions:

1. Even though seabed mining may be economically lucrative, what are two of the biological concerns related to such largescale disturbance of the seafloor?
2. What type of sediments are the deep-sea miners interested in and why?
3. Are mid-ocean ridges the only deep-ocean feature where sulfite deposits form? Based on the Plate Tectonics class, where can we expect such minerals to form?
4. Can we expect to find the valuable minerals over the continental shelf? Why?